WACV #*****

Rethinking Decoder Redundancy in Efficient 3D Medical Image Segmentation: A Systematic Characterization Study

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Abstract

The decoder is an essential module for fusing multi-scale features and restoring spatial detail in U-Net architectures for 3D medical image segmentation. Recent designs, especially attention-based decoders, sharply increase its cost, and efficient methods such as EffiDec3D respond by shrinking high-resolution decoder stages—removing $> 85\%$ *of its parameters and FLOPs across 3D UX-Net, MedNeXt-M-K3 and SwinUNETR with little loss in Dice. This success, however, rests on the empirical assumption that decoder computation is static and globally redundant; it does not explain* why *the computation is removable,* where *the decoder still matters, or whether those regions can be identified at test time. We present a systematic characterization of decoder redundancy along three axes—*heterogeneity*,* concentration*, and* predictability*—by counting per-voxel prediction* flips *between matched full and efficient decoders, with a frozen-backbone control and a channel/resolution factor decomposition that isolate the decoder path. We find the full decoder's net benefit is small and* bidirectional to the point of cancellation*: at the parameter-matched configuration the net flip rate is statistically indistinguishable from zero across matched backbones (e.g.* -0.001% *on SwinUNETR)—yet this voxel-level neutrality coexists with a* 1–2 *point Dice gap, because the balanced flips concentrate at the Dice-weighted organ boundaries. Uncertainty reliably predicts* where *the full decoder changes a prediction (AUROC* ≈ 0.9*) but not* whether *the change improves it: predicting instability is easier than predicting improvement. These trends are consistent across architectures, indicating that the removed decoder capacity contributes almost nothing to net voxel correctness while still influencing macro metrics through a thin boundary band. Our analysis provides an evaluation protocol and design guidance for more efficient, region-adaptive segmentation decoders.*

1. Introduction

Encoder–decoder U-Nets are the dominant architecture for 3D medical image segmentation, where the decoder fuses multi-scale encoder features and restores the spatial detail needed for accurate organ boundaries. As backbones have grown—from convolutional [9, 13] to Transformer [6, 7, 14] designs—the decoder has become a large share of the compute and memory budget, particularly its high-resolution stages that operate at or near input resolution. Efficient-decoder methods respond by shrinking these stages: EffiDec3D, a recent example, removes more than 85% of the decoder's parameters and FLOPs across 3D UX-Net, MedNeXt-M-K3, and Swin UNETR while losing little Dice [12].

Such results demonstrate that decoder computation is *removable*: one can delete it and watch aggregate accuracy hold. They do not, however, explain *why* it is removable, *where* in the volume the full decoder still changes predictions, or whether those regions can be identified at test time. This gap matters for the field's next step. If the removed capacity were uniformly useless, efficient decoders could stop at a fixed reduction; if instead its benefit were concentrated and predictable, one could allocate decoder computation *adaptively*, spending it only where it helps. Aggregate Dice and FLOPs cannot distinguish these cases, because a metric that averages over millions of voxels hides both the spatial structure of the change and its direction.

We therefore complement the ablation evidence with a systematic characterization of decoder redundancy. Rather than asking whether the decoder can be removed, we ask *how* its contribution is distributed, organizing the study along three axes: **heterogeneity** (is the full decoder's effect uniform, or does it depend on location and anatomy?), **concentration** (is the beneficial computation confined to a small subset of regions?), and **predictability** (can a test-time signal find those regions without the ground truth?). Our unit of analysis is the per-voxel *prediction flip* between a matched full and efficient decoder—the standard tool for comparing two classifiers on paired data [2, 10, 15] —which

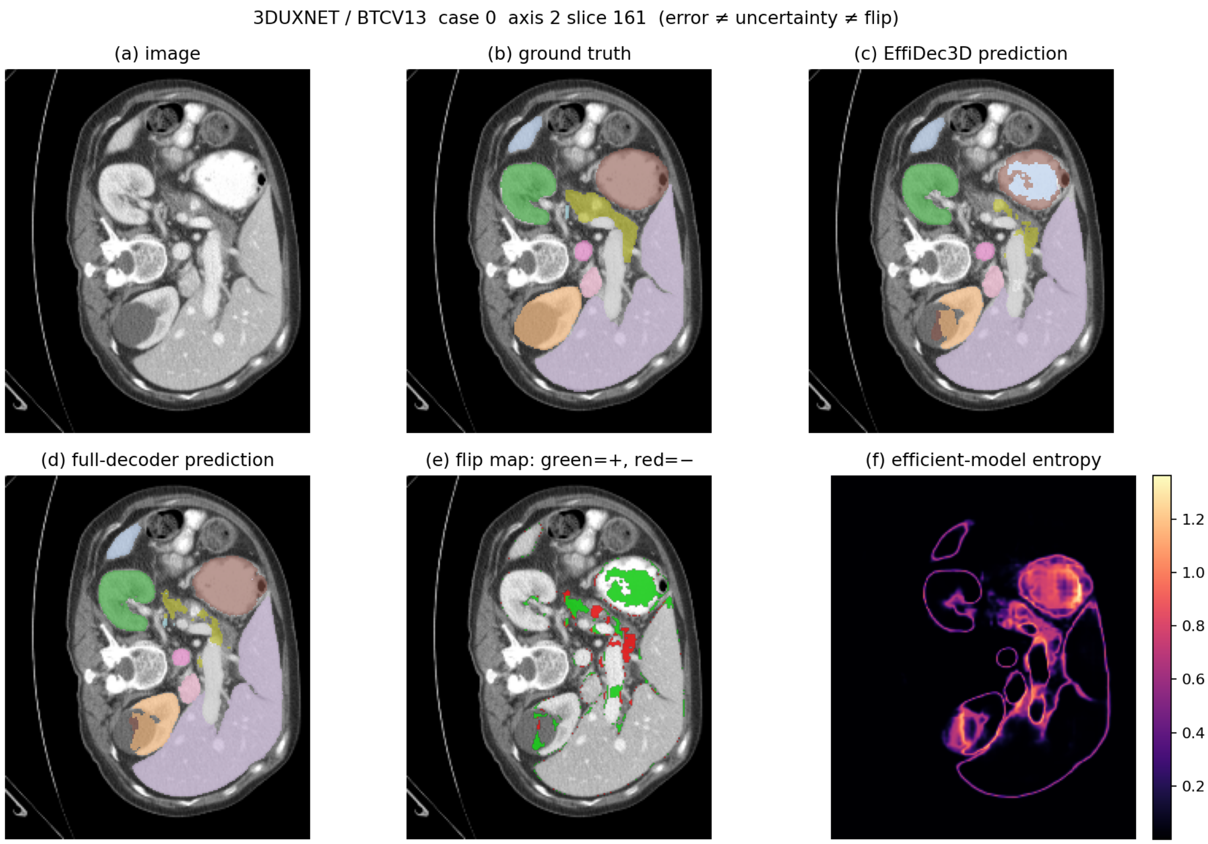

Figure 1. **Decoder-induced prediction changes are spatially localized but bidirectional.** For one input volume (a) with ground truth (b), the efficient (c) and full (d) decoders differ primarily near anatomical boundaries. The flip map (e) shows the full decoder *corrects* some efficient-model errors (green, positive flips) while simultaneously *introducing* new errors (red, negative flips); both overlap the efficient model's high-entropy regions (f). The zoomed positive (g) and negative (h) examples share the same high uncertainty, illustrating our central distinction: predicting *where* a prediction will change is easier than predicting *whether* the change is beneficial. This figure is a qualitative teaser; the quantitative evidence is in Sec. 3.

separates cases where the full decoder *corrects* the efficient one (positive flips) from those where it *degrades* it (negative flips). We evaluate two regimes: an end-to-end matched pair, and a frozen shared-encoder control that isolates the decoder path. Generality is tested with an L-shaped protocol spanning three architectures (3D UX-Net, Swin UNETR, MedNeXt-M-K3) and three datasets (BTCV [1], FeTA [11], and MSD HepaticVessel).

Our characterization yields a nuanced picture that the aggregate numbers obscure. At the parameter-matched configuration the full decoder's net benefit is *bidirectional to the point of cancellation*: it corrects and degrades comparable numbers of voxels, so the net flip rate is statistically indistinguishable from zero (e.g. -0.001% on Swin UNETR). Yet this voxel-level neutrality coexists with a 1–2 point Dice gap, because the near-balanced flips concentrate in a thin band within one to two voxels of organ boundaries—exactly where Dice is most sensitive. Finally, the efficient model's own uncertainty predicts *where* a prediction will change (AUROC ≈ 0.9) but not *whether* the change is an improvement (direction AUROC ≈ 0.5): predicting instability is easy, predicting improvement is not. This last result is the central obstacle to turning decoder redundancy into a deployable, benefit-aware efficient decoder, and it holds consistently across the three architectures.

Contributions.

- We reframe efficient-decoder evaluation from "can the computation be removed?" to "where and why is it removable?", and give an evaluation protocol—prediction flips resolved by boundary distance and anatomy, an oracle concentration analysis, and flip-prediction AUROC—that exposes what aggregate Dice and FLOPs hide.
- We characterize the redundancy empirically: the full decoder's net voxel benefit is statistically indistinguishable from zero yet Dice-relevant through a thin boundary band, and this net-neutral-but-boundary-localized pattern

Table 1. L-shaped protocol. The 3D UX-Net row tests cross-dataset consistency; the BTCV column tests cross-architecture consistency; they meet at the UX-Net/BTCV anchor ($*$, completed). Off-axis cells (–) are not run.

Backbone	BTCV (CT)	FeTA (MRI)	HepaticVessel (CT)
3D UX-Net	✓*	✓	✓
Swin UNETR	✓	–	–
MedNeXt-M-K3	✓	–	–

is consistent across three architectures and three datasets.

- We show that the beneficial flips are predictable in *location* but not in *direction*, quantify the resulting ceiling on test-time region selection, and draw concrete implications for designing and auditing region-adaptive decoders.

2. Experimental Setup

Experiment plan. Our evaluation study reproduces EffiDec3D [12] and evaluates it with an *L-shaped* protocol (Table 1) that separately tests cross-dataset and cross-architecture consistency without an expensive full factorial. A **dataset axis** runs the matched 3D UX-Net [9] pair on three diverse tasks from the EffiDec3D evaluation—BTCV abdominal CT, FeTA [11] fetal-brain MRI, and MSD Task08 HepaticVessel [1] (CT, thin vessels/small lesions)—and an **architecture axis** runs matched pairs on BTCV for three backbone families: 3D UX-Net (large-kernel CNN), Swin UNETR [14] (hierarchical Transformer), and MedNeXt-M-K3 [13] (ConvNeXt-style CNN). The two axes share the 3D UX-Net/BTCV anchor, which also carries the mechanism analysis below.

Dataset and split. The primary task is the 13-organ BTCV/Synapse abdominal CT benchmark with 18 training and 12 validation volumes: spleen, right and left kidney, gallbladder, esophagus, liver, stomach, aorta, inferior vena cava, portal and splenic veins, pancreas, and right and left adrenal glands. Images are intensity-windowed and resampled following the public EffiDec3D pipeline; the same validation subjects are paired across all model comparisons on a given dataset.

Two comparison regimes. Within each cell the full and EffiDec3D models share the same encoder *architecture* and differ in the decoder (channels reduced to the minimum encoder width, additive skip aggregation, resolution factor two). We analyze them under two regimes. (i) **End-to-end (ecological).** Both models are trained independently, so they share the encoder architecture but not its weights; their flips reflect the deployed full-vs-efficient difference but conflate decoder capacity with encoder-weight and optimization differences. (ii) **Frozen shared encoder (causal).** We take one trained encoder, freeze it, and train the decoder variants on *identical* features, so prediction differences are attributable to the decoder alone. Both regimes hold the encoder architecture fixed, so the study characterizes the decoder and draws no conclusions about the encoder itself.

Frozen-encoder factor decomposition. EffiDec3D removes decoder computation in two ways at once—narrower channels and a dropped high-resolution stage. To attribute effects, we decompose them with a 2×2 factorial on the frozen encoder, all within the EffiDec3D decoder family so only the two knobs vary: full (highest-resolution stage, wide channels), channel-only (wide $\rightarrow$ 48 channels), resolution-only (drop the high-resolution stage), and the combined EffiDec3D decoder. Pairwise flips (Full$\rightarrow$channel-only, Full$\rightarrow$resolution-only, Full$\rightarrow$combined) isolate which factor changes which regions. The frozen encoder is optimized for the full decoder, so "efficient still matches on frozen features" is a *conservative* redundancy result.

Null-pair seed control. Because the net flip is tiny, we bound seed noise with a same-architecture null pair: two Full models trained with different (predeclared) seeds. The claim is that Full-vs-EffiDec3D net flips and their concentration exceed this Full-vs-Full null.

Implementation. We reproduce the EffiDec3D training protocol on a single NVIDIA RTX 5090 (32 GB, Blackwell) under PyTorch 2.6 / CUDA 12.8 with MONAI, using BF16 mixed precision (no loss scaling). Each model is trained for 45,000 optimizer steps (validation every 250) with AdamW at learning rate 10^{-3}, a Dice–cross-entropy loss, four 96^3 random-crop samples per volume, and full in-memory caching. Sliding-window inference uses a 96^3 window, batch size four, overlap 0.7, and constant blending. We evaluate one full run and two independently initialized EffiDec3D runs per cell to expose seed sensitivity. Segmentation quality is reported with Dice and 95th-percentile Hausdorff distance (HD95), and efficiency with parameters, multiply–accumulate operations (MACs), latency, and peak memory.

Core quantities. Let $\hat{y}_{\mathrm{E}}(v)$, $\hat{y}_{\mathrm{F}}(v)$, and $y(v)$ be the efficient prediction, full prediction, and ground-truth label at voxel v. We measure where the full decoder changes predictions with *prediction flips*, the standard tool for comparing two classifiers on paired data [2, 15]:

$$P(v) = \mathbb{1}[\hat{y}_{\mathrm{E}}(v) \neq y(v) \ \wedge \ \hat{y}_{\mathrm{F}}(v) = y(v)]\,, \tag{1}$$
$$N(v) = \mathbb{1}[\hat{y}_{\mathrm{E}}(v) = y(v) \ \wedge \ \hat{y}_{\mathrm{F}}(v) \neq y(v)]\,, \tag{2}$$
$$U(v) = P(v) - N(v), \tag{3}$$

a *positive* flip (full corrects efficient), a *negative* flip (full degrades efficient), and the *net* flip. P and N are exactly the discordant cells of the paired-agreement table behind McNemar's classifier-comparison test [3, 10]; volume rates are $R_{\mathrm{pos}} = \mathrm{mean}_v P(v)$, $R_{\mathrm{neg}} = \mathrm{mean}_v N(v)$, and $R_{\mathrm{net}} = R_{\mathrm{pos}} - R_{\mathrm{neg}}$. As test-time signals computable from the efficient model alone we use predictive entropy $H(v) = -\sum_c p_c(v) \log p_c(v)$ and confidence $\max_c p_c(v)$.

Characterization metrics. We organize the analysis as three questions and report exactly six metrics; each is computed per grid cell and aggregated over subjects with a bootstrap, since the twelve validation volumes—not the correlated voxels—are the independent units.

(1) Heterogeneity: is the decoder's effect uniform, or does it depend on location and structure?

- **H1, global flip rates.** The subject-mean R_{pos}, R_{neg}, and $R_{\mathrm{net}} = R_{\mathrm{pos}} - R_{\mathrm{neg}}$ of Eqs. (1)–(3), with a subject bootstrap CI on R_{net}. *Meaning:* whether the full decoder improves predictions uniformly or simultaneously corrects and degrades them—$R_{\mathrm{net}} \approx 0$ with $R_{\mathrm{pos}}, R_{\mathrm{neg}} > 0$ signals a bidirectional cancellation that a Dice gap alone cannot reveal.
- **H2, boundary- and anatomy-resolved net flips.** $R_{\mathrm{net}}(D_k)$ over voxels binned by Euclidean distance $d(v)$ to the nearest GT boundary ($D_k \in \{0\text{–}1, 1\text{–}2, 2\text{–}4, >4\}$), and, on the $\mathrm{union}(\mathrm{GT}, \mathrm{Full}, \mathrm{Effi})$ foreground of each organ c (so false positives are not dropped), $R_{\mathrm{net},c}$ alongside the per-organ Dice change $\Delta\mathrm{Dice}_c = \mathrm{Dice}_{\mathrm{F},c} - \mathrm{Dice}_{\mathrm{E},c}$ (NSD at 1–2 voxel tolerance as an optional boundary-quality column). *Meaning: where* the effect lives—whether the net benefit concentrates in a thin boundary band and in particular structures, explaining how a near-zero global net coexists with a macro Dice gap.

(2) Concentration: is the beneficial computation spatially concentrated?

- **C1, oracle net-benefit coverage.** Partition each volume into contiguous 16^3 blocks, score block r by its true net benefit $B(r) = \sum_{v \in r} U(v)$, rank by $B(r)$, and plot the cumulative recovered fraction $C_{\mathrm{oracle}}(k) = \sum_{r \in \text{top-}k} B(r) \,/\, \sum_r \max(B(r), 0)$ against the block budget k (normalizing by the positive net mass, not all positive flips). *Meaning:* the ceiling—whether, *with oracle knowledge*, a small region subset carries the net benefit. We report the absolute net $\sum_r B(r)$ beside the coverage so a steep curve over a near-zero total is not mistaken for a large opportunity.
- **C2, top-k coverage.** The oracle coverage at 10% and 20% of blocks and the smallest budget K_{80} reaching 80% of the net benefit. *Meaning:* a compact summary of *how* concentrated the opportunity is.

(3) Predictability: can a test-time signal find the beneficial regions without the ground truth?

- **P1, flip-prediction AUROC.** Subject-level AUROC of the efficient model's entropy $H(v)$ [8] for three progressively harder targets: *any* correctness-changing flip $P \vee N$, *positive* flips P, and— restricted to flip voxels—the *direction* P vs. N. *Meaning:* whether uncertainty predicts *where* predictions change, *where* they are corrected, and *whether* a change is an improvement; a direction AUROC near 0.5 means instability is predictable but improvement is not.
- **P2, deployable region selection.** Using the same blocks, rank by a ground-truth-free signal (entropy, confidence, a boundary proxy) and compare the recovered net benefit against the oracle and a matched-random selector [4], reporting the paired (signal−random) gap with a subject bootstrap CI at the 20% budget. *Meaning:* whether the concentrated opportunity C1 identifies is actually *recoverable* at inference—a signal helps only if its gap over random excludes zero.

Cross-dataset (**O7**) and cross-backbone (**O8**) consistency re-evaluate these six metrics across the rows and columns of Table 1. Error-based diagnostics—boundary error ratio, per-organ difficulty, entropy sparsity, and the halo/executed-volume opportunity variant—are reported only in the appendix.

[TODO: Completed results below are the 3D UX-Net/BTCV grid cell. Remaining cells (other backbones and datasets) are in progress; cross-backbone (columns) and cross-dataset (rows) consistency are reported once enough cells are filled.]

3. Results

3.1. Controlled Reimplementation

We reproduce the matched Full-UX/Effi-UX pair and report its fidelity per dataset (Table 2). On BTCV our Effi-UX reaches 0.777 mean Dice, 1.5 points below the published 0.793, while Full-UX is within 0.6 points of its published value; with EffiDec3D's default concatenation skip our Effi-UX matches its published 3.15 M parameter count, so the residual gap reflects our reconstructed identity-affine data rather than the paper's processed set (a FP32-versus-BF16 check ruled out a numerical cause). Efficiency reproduces the published reductions: Effi-UX uses 11.7× fewer MACs, 16.8× fewer parameters, ∼5× lower latency, and ∼ 7.7× lower peak memory. The same pattern holds

Table 2. Reproduction fidelity of the matched 3D UX-Net pair, per dataset. We report mean Dice of our runs against the published EffiDec3D value; Δ is our Effi-UX minus the published Effi-UX. Data were reconstructed with identity affine metadata, so this is a controlled reimplementation, not an exact reproduction.

Dataset	Full-UX	Effi-UX	pub. Effi-UX	Δ
BTCV (CT, 13-organ)	.792	.777	.793	−.016
FeTA (MRI)	–	–	–	–
HepaticVessel (CT)	–	–	–	–

[TODO: Fill FeTA/HepaticVessel rows as the dataset-axis cells complete.]

Table 3. Architecture axis (BTCV): reproduction and efficiency of the matched full/EffiDec3D pair per backbone. Dice is Full→EffiDec3D; reductions are Full/EffiDec3D.

Backbone	Dice (F→E)	MAC↓	Param↓	Lat↓
3D UX-Net	.792→.777	11.7×	16.8×	5.1×
Swin UNETR	.805→.795	5.5×	5.5×	2.2×
MedNeXt-M-K3	.837→.815	2.2×	3.0×	2.2×

on the architecture axis (Table 3): on BTCV, Swin UNETR reproduces its published full model (0.805 vs. 0.8013) and its EffiDec3D decoder cuts MACs 6.5× at a 2.6-point Dice cost—a smaller reduction than 3D UX-Net's, matching EffiDec3D's own per-backbone figures. MedNeXt-M-K3, the strongest backbone at 0.837 full Dice, shows a 2.3-point cost at a 2.2× MAC reduction; across the three backbones the Dice cost is small (1.0–2.3%) even as the achievable MAC reduction ranges from 2.2× to 11.7×—consistent across architectures, not universal in magnitude. Treating these matched models as fixed, the rest of the paper characterizes *where* the full decoder's extra computation changes predictions—the question EffiDec3D's aggregate evaluation leaves open.

3.2. Heterogeneity Metrics

With the parameter-matched Effi-UX, the full decoder's benefit is small and *bidirectional to the point of cancellation*. On 3D UX-Net, Full-UX corrects 0.278% of Effi-UX's voxels (R_{pos}) while degrading 0.232% (R_{neg}), so the net flip rate R_{net} is only 0.047% and its subject-level 95% CI $[-0.026\%, +0.112\%]$ *includes zero* (Table 4); on Swin UNETR the two rates are equal (0.201% each) and R_{net} is −0.001%. Across both matched backbones the full decoder's net benefit is thus *statistically indistinguishable from zero at the voxel level*. Crucially, this net neutrality coexists with the 1–2 point Dice gap of Table 3: the flips, though nearly balanced in count, concentrate at organ boundaries that Dice weights heavily, so the full decoder still moves macro metrics through a thin boundary band

Table 4. Prediction-flip rates for the parameter-matched (concatenation) efficient decoder (fractions of evaluated voxels). Positive/negative flips are the discordant cells of the paired comparison [10, 15]. The net-flip CI includes zero.

Metric	value
Positive flip rate R_{pos}	.00278
Negative flip rate R_{neg}	.00232
Net flip rate R_{net}	.00047
Net flip rate, subject 95% CI	[−.00026, .00112]

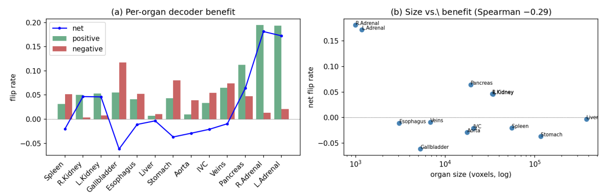

Figure 2. H2, per-organ net flip for the matched pair (seed 1): positive, negative, and net rates on each organ's union(GT, Full, Effi) foreground, and net flip vs. organ size. The decoder's net benefit is uneven across anatomy and rises as structures shrink.

even as its net contribution to voxel correctness vanishes. Reporting positive flips alone would overstate a voxel-level benefit that, netted against its own regressions, is essentially none.

This near-zero global net is *not* a uniform absence of effect but a spatially structured cancellation (H2). Resolving R_{net} by distance to the GT boundary shows the benefit is sharply boundary-localized: it peaks at the surface (0.085 at 0–1 vox) and falls to near zero or slightly negative beyond one voxel [TODO: refresh H2 boundary profile from re-run], so the balanced flips of Table 4 live almost entirely in a one-voxel shell—the same band Dice weights heavily. Per organ (on the union foreground), the net benefit rises as structures shrink (adrenals, pancreas, veins; organ-size/net-flip $\rho = -0.37$, Fig. 2), with the per-organ Dice change ΔDice_c tracking the same small structures; surface Dice (NSD) is 0.741/0.845 at τ=1/2 mm. This boundary and small-structure concentration should exceed a same-architecture null pair (Full seed 0 vs. seed 1) [TODO: null-pair floor], ruling out seed noise. (The error-based boundary ratio—Effi-UX error 4.39× the interior rate—is a consistent but weaker error-side view, reported in the appendix.)

3.3. Concentration Metrics

We next ask whether that small net benefit is at least spatially concentrated (C1). Ranking contiguous 16^3 blocks by their true net benefit $B(r) = \sum_{v \in r} U(v)$, the top 10%/20% of blocks recover [TODO: C_{10}/C_{20}%] of the volume's net benefit, and K_{80} = [TODO:]% of blocks are needed to reach 80% (Fig. 3, oracle curve). The coverage curve is steep, so the net benefit *is* concentrated—but it is concen-

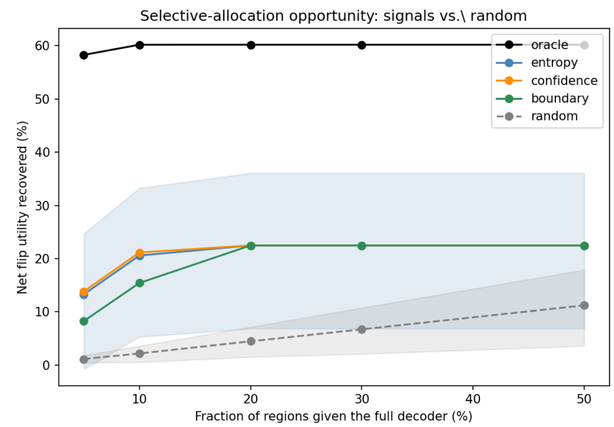

Figure 3. C1/P2: net benefit recovered vs. the fraction of 16^3 blocks selected. The true-net *oracle* (black; C1) upper-bounds concentration, while deployable signals (entropy, confidence, boundary) and a matched-random selector (P2) show whether that opportunity is recoverable at test time. Shaded bands are subject 95% CIs.

Table 5. P1: subject-level AUROC of efficient-model entropy for three progressively harder flip-prediction targets. Entropy locates *where* predictions change but carries almost no information about *whether* they improve (direction ≈ 0.5).

Target	AUROC	95% CI
Any flip ($P \vee N$)	[TODO:]	[TODO:]
Positive flip (P)	.903	[.876, .926]
Direction (P vs. N)	[TODO: $\approx$ .5]	[TODO:]

trated over a total that is itself ≈ 0 (Table 4), a small absolute opportunity rather than a large one. We report coverage and absolute net together precisely so the curve's shape is not read as a large payoff. For reference, a voxel-level oracle over entropy captures 83.1% of the *positive* flips in its top 20%; but positive-flip coverage ignores the near-equal negative flips that cancel them, which is why the net-benefit oracle above is the honest concentration measure.

3.4. Predictability Metrics

The scarce flips, though net-neutral, are not random—but only their *location*, not their *direction*, is predictable from the efficient model's own uncertainty (P1, Table 5). Entropy separates any correctness-changing flip from stable voxels (AUROC [TODO:]) and predicts the *positive*-flip voxel with per-subject AUROC 0.903 (95% CI $[0.876, 0.926]$) [8]. Restricted to the flip voxels themselves, however, entropy cannot tell an improvement from a regression: the positive-vs-negative (direction) AUROC is [TODO: $\approx$ 0.5], so knowing that a prediction will change conveys almost nothing about whether it gets better. Predicting *instability* is easy; predicting *improvement* is not—the central obstacle to turning this redundancy into a benefit-aware efficient decoder. (Entropy is moderately calibrated, ECE 0.088, $[0.065, 0.115]$ [5]; the pooled-bin entropy–error correlation $r = 0.97$ is a descriptive diagnostic only.)

This unpredictable direction directly bounds deployable region selection (P2). Using the same 16^3 blocks as C1 but ranking by a ground-truth-free signal [4], we compare the net benefit recovered against the oracle and a matched-random selector under a paired subject bootstrap (Fig. 3). Because the net benefit is ≈ 0 and its direction is unpredictable, recovery is weak: at the predeclared 20% block budget the entropy$-$random gap is [TODO: $g\pm$CI] with a 95% CI that includes zero [TODO: confirm from re-run], and on Swin UNETR the paired lower bound is negative at every budget. We therefore report a *negative* result at the matched configuration: entropy localizes where the scarce flips occur, but there is no reliable net benefit to selectively recover. Entropy and confidence are interchangeable as signals (paired difference ≤ 0.02), so we do not claim entropy is uniquely optimal.

3.5. Mechanism: Which Factor Is Removable

The comparisons above use end-to-end models, which share the encoder architecture but not its weights. As a further, causally cleaner check, we *freeze* one trained encoder and train the 2×2 decoder factorial on identical features, then compare Full→channel-only, Full→resolution-only, and Full→combined EffiDec3D [TODO: report per-factor flip rates and boundary profiles from the frozen-encoder run; hypothesis to test: dropping the high-resolution stage yields flips concentrated nearer the boundary, while channel reduction is spatially diffuse and smaller, which would explain why both are removable with little aggregate Dice change]. Because the frozen encoder was optimized for the full decoder, a near-match by the efficient decoder on those features would be a *conservative* statement of redundancy. We present this regime as validation of the decoder attribution rather than as the primary result.

4. Discussion: Implications for Efficient Decoder Design

Our characterization has three implications for how efficient 3D decoders are built and evaluated. **First, aggregate Dice and FLOPs are insufficient summaries.** They hide that the full decoder's advantage over a parameter-matched efficient decoder is, at the voxel level, a near-perfect cancellation of improvements and regressions ($R_{\text{net}} \approx 0$): a decoder can differ by $\sim$ 1–2 Dice points while providing no net voxel-level benefit. Reporting positive, negative, and net flips exposes this and should accompany Dice/FLOPs whenever a decoder is simplified. **Second, the redundancy is spatially structured, not diffuse.** The flips that remain concentrate within one to two voxels of organ boundaries and on small structures, and are predictable from the efficient model's own entropy (AUROC ≈ 0.9); efficient decoders can thus be designed and audited around a small, identifiable boundary band rather than the whole volume. **Third, the redundancy is a property of the decoder, not a single backbone.** Across 3D UX-Net, SwinUNETR, and MedNeXt the same pattern holds—small Dice cost, boundary-localized predictable flips, and a voxel-level net benefit statistically indistinguishable from zero—indicating that the decoder stages removed by EffiDec3D contribute almost nothing to *net voxel correctness*, even though, via that same boundary band, they still account for a 1–2 point macro-Dice gap. EffiDec3D's aggressive compression is thus empirically well justified, and the removed capacity is better described as net-neutral than as strictly unused. Future efficient decoders should explicitly measure *where* computation changes predictions rather than assume a uniform accuracy cost.

References

[1] Michela Antonelli, Annika Reinke, Spyridon Bakas, et al. The medical segmentation decathlon. *Nature Communications*, 13(1):4128, 2022. 2, 3

[2] Quentin Cormier, Mahdi Milani Fard, Kevin Canini, and Maya R. Gupta. Launch and iterate: Reducing prediction churn. In *Advances in Neural Information Processing Systems*, 2016. 1, 4

[3] Thomas G. Dietterich. Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation*, 10(7):1895–1923, 1998. 4

[4] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, 2017. 4, 6

[5] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning*, pages 1321–1330, 2017. 6

[6] Ali Hatamizadeh, Yucheng Tang, Vishwesh Nath, Dong Yang, Andriy Myronenko, Bennett Landman, Holger R. Roth, and Daguang Xu. Unetr: Transformers for 3d medical image segmentation. In *IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 1748–1758, 2022. 1

[7] Yufan He, Vishwesh Nath, Dong Yang, Yucheng Tang, Andriy Myronenko, and Daguang Xu. SwinUNETR-V2: Stronger swin transformers with stagewise convolutions for 3d medical image segmentation. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 416–426, 2023. 1

[8] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations*, 2017. 4, 6

[9] Ho Hin Lee, Shunxing Bao, Yuankai Huo, and Bennett A. Landman. 3d ux-net: A large kernel volumetric convnet modernizing hierarchical transformer for medical image segmentation. In *International Conference on Learning Representations*, 2023. 1, 3

[10] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947. 1, 4, 5

[11] Kelly Payette, Priscille de Dumast, Hamza Kebiri, et al. An automatic multi-tissue human fetal brain segmentation benchmark using the fetal tissue annotation dataset. *Scientific Data*, 8(1):167, 2021. 2, 3

[12] Md Mostafijur Rahman and Radu Marculescu. Effidec3d: An optimized decoder for high-performance and efficient 3d medical image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10435–10444, 2025. 1, 3

[13] Saikat Roy, Gregor Koehler, Constantin Ulrich, Michael Baumgartner, Jens Petersen, Fabian Isensee, Paul F. Jaeger, and Klaus H. Maier-Hein. MedNeXt: Transformer-driven scaling of ConvNets for medical image segmentation. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 405–415, 2023. 1, 3

[14] Yucheng Tang, Dong Yang, Wenqi Li, Holger R. Roth, Bennett Landman, Daguang Xu, Vishwesh Nath, and Ali Hatamizadeh. Self-supervised pre-training of swin transformers for 3d medical image analysis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20730–20740, 2022. 1, 3

[15] Sijie Yan, Yuanjun Xiong, Kaustav Kundu, Shuo Yang, Siqi Deng, Meng Wang, Wei Xia, and Stefano Soatto. Positive-congruent training: Towards regression-free model updates. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14299–14308, 2021. 1, 4, 5